

# Nguyen Van Kinh

AI Engineer

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## OBJECTIVE

Data Science graduate (University of Science – VNU-HCM) and AI Engineer building LLM-powered AI agents, multi-agent systems, and workflow automation. Experienced in taking agentic systems from prototype to production — conversational voicebots, RAG and knowledge-graph assistants, and multi-agent data platforms — with a strong emphasis on evaluation, monitoring, and continuous improvement.

## EDUCATION

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|---------------------------------------------------------------------|--------------------|
| University of Science, Vietnam National University Ho Chi Minh City | 2022 – 2026        |
| Bachelor of Science in Data Science — <b>Excellent</b>              | GPA: <b>3.83/4</b> |

## WORK EXPERIENCE

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|--------------------|--------------------|
| Junior AI Engineer | Feb 2026 – Present |
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[Namitech](#)

### Project: NamiGen – Smart IVR & Conversational AI Platform (Voicebot/Chatbot)

Multi-tenant, config-driven voicebot platform delivering smart IVR, FAQ, and call-transfer assistants in Japan and Vietnam from a single codebase.

#### Responsibilities:

- **Intent routing:** Owned the LLM intent router of a Japanese city-hall voicebot, now in production (363 calls in its first 3 hours). Callers speak in fragments, with ASR errors and subject-less Japanese. Over 4 eval-driven rounds on a 618-utterance test set, raised accuracy from **64% to 99%** through sharper intent boundaries and contrastive few-shots, while cutting the prompt by **30%** (30k → 21k tokens).
- **FAQ answering (RAG):** Rebuilt FAQ answering on hybrid retrieval (Qdrant) with reranking; the bot reads the approved answer verbatim, with numbers and times rewritten for TTS. CJK-aware grounding stopped valid Japanese answers from being rejected, and a *no-evidence* → *transfer* fallback keeps the bot from inventing answers. Raised the pass rate on ~1,800 test questions from **46% to 98%**, with <1% wrong answers.
- **Smart IVR & call transfer:** Replaced keypad menus with spoken routing: a directory matcher maps what the caller says to the right one of **634 departments** (Unicode, kanji and full-width digit normalization), with L1/L2 policy (bot answers vs. operator transfer) and business-hours, night/holiday, and emergency branches.
- **Dialog engine & LLM layer:** Extended the graph-based dialog engine with structural go-back/repeat, jump-to-step, clarify/re-clarify escalation, and silence/wait-more handling. Added Amazon Bedrock (Nova) support with schema-forced JSON, throttling retries, and reasoning-model fallbacks.
- **AI recruiting:** Built a screening-interview voicebot and a STAR scoring API for an insurer in Indonesia (parallel rubric scoring, versioned prompts in MongoDB), plus a weekly hiring-report API whose figures are computed in code and checked against the LLM's text before release.
- **Delivery & latency:** Shipped voicebots for **10+ clients** in Vietnamese, Japanese and English at **1.3–2s TTFT** (end of caller speech to first bot audio). All customer behavior lives in config, so a new PoC ships in **1–3 weeks** instead of **1–2 months**.

#### Projects:

- **Japan:** Sagamihara City smart IVR (**production**); Osaka City, Shidax, CBA (smart IVR/FAQ PoCs).
- **Other PoCs:** REI (JP bank card-lock line), Rinnai (JP hands-free cooking assistant), BIDV (VN banking fraud/complaint line), PVI (VN insurance ticketing).

**Tech Stack:** Python, FastAPI, WebSocket, MongoDB, Qdrant, Redis, Docker, Langfuse, DeepEval, Azure OpenAI, Amazon Bedrock, CI/CD.

*MetaData Solutions***Project: 1Brain – Data Intelligence Platform**

Multi-agent platform turning unstructured documents into graph data for querying and exploration.

**Responsibilities:**

- Designed and implemented a multi-agent architecture with graph-based retrieval that **improved query accuracy from 66% to 85%** versus native RAG, processing **10–20 concurrent documents** (15–30 pages each) into structured knowledge graphs for complex querying and reasoning.

**Tech Stack:** LangChain, LangGraph, LangSmith, FastAPI, FalkorDB, PostgreSQL, RAG.

**Project: Telehealth – Chatbot Medical Assistant for Doctors**

Knowledge graph-powered assistant that structures medical records and automates admin actions.

**Responsibilities:**

- **Improved chatbot accuracy from 62% to 83% (F1-score)** by grounding responses in clinically verified graph data, supporting complex medical queries.
- Built agent tools for CRUD on patient databases (real-time record updates, appointment scheduling) and a Graphiti pipeline turning medical Google Docs (text and images) into a patient-centric FalkorDB knowledge graph.

**Tech Stack:** LangChain, Graphiti, FalkorDB, FastAPI, LangFuse, RAG.

**PERSONAL PROJECTS**

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**Multi-Agent System for Multimedia Soccer Knowledge Understanding** | [GitHub](#)

*Role: AI Engineer / Core Developer*

Multi-agent conversational AI embedded into soccer streaming platforms, enabling real-time news/web search and live in-broadcast Q&A by orchestrating structured, vector, and cache stores with multi-worker planning and async media processing.

**Key Features & Architecture:**

- **Multi-Agent Orchestration:** LangGraph DAG routing with PostgreSQL checkpointing for crash recovery.
- **Multi-Intent Planning:** planner resolves co-references via **Qdrant** few-shots, splitting questions into parallel tool calls.
- **Multi-Tier Memory:** filters tool traces, semantic dedup ( $\cosine \geq 0.95$ ), token-aware pruning.
- **Live-Video Indexing:** Celery workers transcribe live HLS streams with **Qwen3-ASR**.
- **Facial Search:** **Qwen-VL** + **ArcFace** embeddings over **gRPC** for player identification.

**Key Performance Outcomes:**

- **SoccerBench SOTA:** cross-task accuracy **69.3% → 75.5%** (peak **96.4%** on text QA).
- **Semantic Caching:** repeated-query latency **–55%** (12.1s → 5.5s p50) via Redis.
- **Parallel Scaling:** a 3-entity image query adds only **0.6s** over a 1-entity query, vs.  $\sim 3\times$  sequentially.

**Technical Stack:** FastAPI, LangGraph, LangChain, Langfuse, Google Gemini, Qwen, PostgreSQL, MongoDB, Qdrant, Redis, OpenCV, FFmpeg, DeepFace.

**AWARDS & ACHIEVEMENTS**

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- **AI Challenge TP.HCM 2025** — Building an intelligent multimodal video search & retrieval system; responsible for the search core engine (FAISS HNSW+SQ8 over  $\sim 2M$  vectors, two-tier recall→rerank pipeline, gRPC service).

**SKILLS**

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- **Generative AI & LLMs:** LangChain, LangGraph, LangSmith, Langfuse, DeepEval, Graphiti, RAG, Multi-agent Systems.
- **Deep Learning & CV:** PyTorch, TensorFlow, Scikit-learn, OpenCV, OCR, Object Detection.
- **Backend & Web:** FastAPI, Flask, Next.js, OpenAPI, WebSocket, CI/CD.
- **Databases & Infra:** FalkorDB, Qdrant, PostgreSQL, MongoDB, Redis, AWS (S3, Bedrock), Docker.